

The cos 2φ modulation on tensor-polarized ⁶Li: projected inclusive and coherent measurements

2026-08-17 (development run 5) · revised 2026-08-26 · polligen (evgen/) + polli_fastsim (fastsim/) · companions: plans/06 (background budget), docs/note_cos2phi_coherent_6Li.md, reports/polarized_li_primer.html, reports/reconstruction_chain_report.html (the reconstructed level: money plots 5R, 7R, 6R) · modification history in Appendix A

This note presents two projected EIC measurements of the cos 2φ azimuthal modulation on a transversely tensor-polarized ⁶Li beam: inclusive DIS, requiring only the scattered electron, and coherent diffraction with the ⁶Li detected intact in its 1⁺ ground state. The modulation is generated by photon double helicity flip and probes gluon transversity [1,2]; it has not been measured on any target.

13.5%

coherent-tag acceptance at IP6 for the 0.20 GeV envelope cut, $\exp(-B \cdot p_{\perp}^2 / T_{\text{cut}})$ at $B = 50 \text{ GeV}^{-2}$ (9–20% for $B \in 40\text{--}60$); the cut is documented for top-rigidity beams and is an optics assumption for ⁶Li, and read as an angular envelope it gives 9% at 99.5 GeV/u. The pots' own aperture, measured in the ePIC geometry, is tighter: $1.4 \times 10^{-2} / 5 \times 10^{-5} / 1.9 \times 10^{-17}$ at 40.8 / 99.5 / 137.5 GeV/u (§4.1).

1.1×10^7 / 1.1×10^8

tagged coherent events in 1 / 10 EIC years (scenario $f_0 = 0.04$, constant 0.20 GeV cut); best-bin $\delta A = 1.9 \times 10^{-3} / 6 \times 10^{-4}$, i.e. 5σ floors at 1.0% / 0.3% modulation. Reconstructed level (angular cut, assumed slot cutout): 2.7×10^6 in year 1; 3.1×10^5 on the measured aperture at low energy, none at mid or top.

0.036

deformation-anchored (a_2) of the tagged sample at $P_{zz} = 0.6$ (band 0.018–0.059; one-sided $\times 0.73$ rate-weighting systematic), scaled from the polarized-deuteron CGC calculation [13]; sign flip vs the deuteron predicted

1 • The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle θ_m with the beam, the inclusive cross section per spin projection m is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1)$$

Sign convention of the tensor-rate terms. The b_1, b_2 terms are written in this program's transcription; in the Hoodbhoy–Jaffe–Manohar convention as applied by HERMES and by Cosyn et al. (W. Cosyn, B. Roldan Tomei, A. Sosa, A. Zec, Eur. Phys. J. A 61 (2025) 83, Eq. 27: $A_{zz} = -(2/3) b_1/F_1$ for the axis along the photon) they carry the opposite sign and the $m = 0$ state has the larger rate for $b_1 > 0$. Only the sign of A_{zz} relative to b_1 , and of the constant term of the spin-state ratio, is affected; nothing in the Δ sector is.

The cos 2φ term requires the virtual photon to flip its helicity by two units, which no spin-1/2 constituent can absorb: neither quarks nor bound nucleons contribute at leading twist, so $\Delta(x, Q^2)$ is purely gluonic and vanishes for a nucleus composed of nucleons alone [1]. At leading order it is generated by the gluon transversity distribution δG through the $O(\alpha_s)$ photon–gluon convolution

$$\Delta(x, Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} \left(\sum_q e_q^2 \right) x^2 \int_x^{1/dy} \delta G(y, Q^2)$$

(in the sign convention of Ref. [25]; conventions differ between papers). The only prior experimental document is a JLab letter of intent without a sensitivity projection [3], and the only moment estimate is the bag-model calculation of §3, which puts the asymmetry at the per-mille level; percent-level values arise only in maximal-transversity ($\Delta_{Tg} = \Delta g$) scenarios for proton–deuteron Drell–Yan, described by their authors as optimistic [5]. The projections therefore span the scenario band $\Delta/F_1 \in 10^{-3}\text{--}10^{-2}$.

The measured amplitude and its inversion back to the structure function are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{\Delta} = -\hat{A} \frac{y^2 D_\phi}{1-y}$$

The fastsim approximation set is $A_{||} = D \cdot g_1 / F_1$ (the y^2 -suppressed term dropped), $g_2 = g_2^{WW}$ and no target-mass corrections, which matter only at high x and low Q^2 : the sweet spots sit at $x \leq 0.14$, where $y^2 = 4M^2 x^2 / Q^2 \leq 0.023$ (below 10^{-2} for $x \leq 0.06$). One finite- y effect is not a labeling matter — the tensor-polarized b_1 – b_3 terms feed the $\cos 2\phi$ amplitude at $O(y^2)$, bounded by $y^2(b_1/F_1) \times 1.15$ — the full Cosyn Eq. (17e) combination, $\approx 6.9\times$ the $y^2 b_1/6$ of the b_1 term alone — which is at most 3% of the Δ amplitude at the sweet spots but exceeds a 10^{-3} scenario outright at $x \approx 0.2$, $Q^2 \approx 1 \text{ GeV}^2$, where the amplitude-versus- x panels extend; it is computable from the same data's A_{zz} (reconstruction-chain report §2). Structure functions are toy parameterizations with PDF-grid (CT18NLO/NNPDFpol11) backends behind one interface, so rates hold to a factor ≈ 1.5 and asymmetry shapes are scenarios by construction; $R = \sigma_L/\sigma_T$ is likewise a toy of the right magnitude, 0.157–0.171 across the acceptance against 0.12–0.34 for the published SLAC/E143 R1998 fit, which sits behind an `r_func` hook. R is the one backend input that moves the physics result: F_2 cancels exactly in the $\cos 2\phi$ amplitude (2×10^{-16}), while $\Delta/F_1 = -2(1+R)\hat{A}$ carries R directly, so the published fit shifts Δ/F_1 by +16.6 / +18.0 / +4.7 / –4.4% at the four sweet spots — a systematic on every Δ quoted here.

2 • Machinery, statistics, and binning

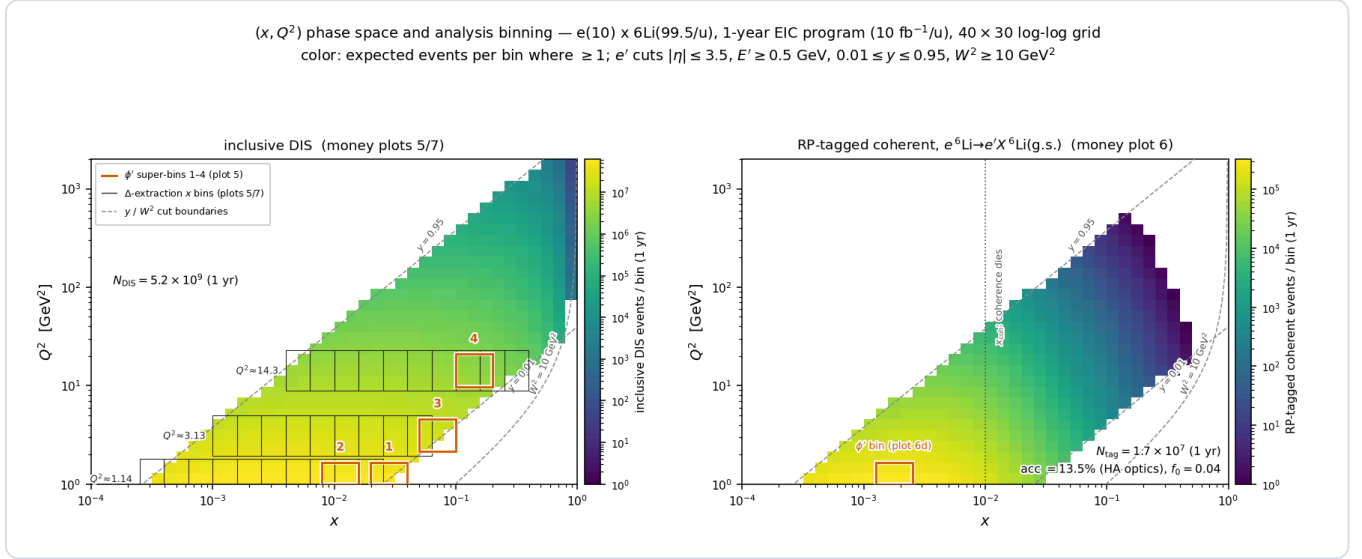
Generator. No public generator produces polarized $e+A$ events for $A > 1$ (BeAGLE is unpolarized; DJANGO and PEPSI are nucleon-level and longitudinal-only), so polligen implements only the missing component — the spin-density matrix $\otimes$ structure-function master formula with spin-correlated sampling — and takes the rest from existing tools, the hadronic final state from a PYTHIA 8 sample of 8 M events over the three beam configurations. Its validation gates are bin-wise identity with the analytic asymmetry formulas at relative tolerance 10^{-12} , pseudo-experiment spreads that close on every analytic error map, and unbiased $\cos 2\phi$ recovery under ϕ -acceptance holes. Since drawing each ϕ -bin count from a Poisson distribution with the exact expected yield is statistically identical to event-level sampling, a full-luminosity pseudo-experiment costs 24 Poisson draws (`sample.phi_histogram_pseudo`); event-level sampling is what the closure tests use.

Estimator of the figures. The amplitudes below come from a single spin state: a binned least-squares fit of $N(\phi') = C(1 + A \cos 2\phi')$ rather than the $2\langle \cos 2\phi' \rangle$ moment, which realistic ϕ holes bias by more than the signal while the fit stays unbiased, with $\delta A = \sqrt{(2/N)/P_{zz}}$ validated by pseudo-experiments and no $\cos \phi'$ term, since the m -symmetric tensor fills set $a_1 = 0$ by construction. The figures below are these single-state, truth-binned projections.

Estimator of the measurement. A single-state fit also measures the detector's own $\cos 2\phi'$ acceptance harmonic divided by P_{zz} — a 3% harmonic on the vertical spin axis fakes four times the signal — so the estimator of the measurement is the spin-state-sorted ratio of $m = \pm 1$ -rich ($P_{zz} = +0.6$) and $m = 0$ -rich (-1.2) bunches (reconstruction-chain report §3). It cancels any acceptance common to both states bin by bin, turns relative-luminosity errors into a constant, and gives $\delta \hat{A} = 2\sqrt{(2/N)/(P_+ - P_0)} = 0.67\times$ the single-state error at equal source purity, provided the states alternate bunch by bunch, or the harmonic is stable to 10^{-4} between fills: a 10^{-3} fill-to-fill difference of it fakes half the signal. At the reconstructed level — EMCal energy, track angles, $\epsilon_{eID}(\eta)$, hadronic-method y , reconstructed (x, Q^2) bins — the four sweet spots give $\delta \hat{A} = 1.2 / 1.0 / 1.8 / 3.0 \times 10^{-4}$ in year 1 (money plot 5R), 0.64–0.70 of the money-plot-5 annotations, and $\Delta(7R)$ reaches $\delta \Delta = 1.0 \times 10^{-3}$ at $Q^2 = 1.14 \text{ GeV}^2$ and 0.5×10^{-3} at 3.13 GeV^2 . Two caveats. With the PYTHIA final state replacing the 25% Gaussian stand-in, sweet-spot purities are 0.40–0.73 rather than 0.64–0.68 and the reconstructed amplitude sits 4–21% below the true bin rather than 1–9%, at unchanged statistical error, on a Σ -method $\delta y/y = 0.55 / 0.28 / 0.50 / 0.15$ at this configuration and 50 MeV of calorimeter noise, degrading with beam energy, which puts the $x \approx 0.1$ bins at low energy; the noise floor decides the $y \approx 0.01$ bins, where $E - p_z$ is 0.2–0.5 GeV and no published ePIC study reaches. Bin-centering is model-dependent at (–5.0, +8.5, –9.3, +6.3)% at the four spots and 3–11% over the plotted bins, above the 10-year statistical errors; since the amplitude is exactly linear in Δ , the response folds as an exact linear operator, and fitting a $\Delta(x)$ shape through it per Q^2 slice reduces that to (–0.9, –1.2, –0.2, +0.3)%.

Binning. The four displayed (x, Q^2) super-bins are selected by statistical significance within Q^2 bands — an unconstrained ranking concentrates at the lowest Q^2 , while the Q^2 lever arm provides the consistency check against power-suppressed

contributions (§6) — and each merges 3×3 cells of the 40×30 analysis grid. Rates use per-nucleon luminosity, F_2^A/A , scattered-electron cuts $|\eta| \leq 3.5$, $E' \geq 0.5$ GeV, $0.01 \leq y \leq 0.95$, $W^2 \geq 10$ GeV², and the full luminosity in the transverse-tensor fill, for the 1-year (10 fb⁻¹/u) and 10-year (100 fb⁻¹/u) programs.



Phase-space coverage and analysis binning (e10 × ⁶Li50, 1-year program, 40×30 log-log grid; color: expected events per bin where ≥ 1; dashed: y and W² cut boundaries). **Left, inclusive DIS** (4.6×10⁹ accepted events in year 1): boxes 1–4 are the ϕ' super-bins of money plot 5; gray combs are the three Q² slices (Q²×[1/1.6, 1.6]) whose merged x-bin pairs give the data points of the amplitude-versus-x panel and of the Δ extraction (plots 5 and 7), retained where $N \geq 10^3$ and $\delta A \leq 8 \times 10^{-3}$. **Right, RP-tagged coherent channel**: the same map times $f_{\text{coh}}(x)$ and the 13.5% tag acceptance, $N_{\text{tag}} = 1.1 \times 10^7$ in year 1, 98% of it at $x < 0.03$ (median 3.5×10^{-3}); the box is the super-bin of money plot 6(d), and plot 6(c) is this map summed over Q².

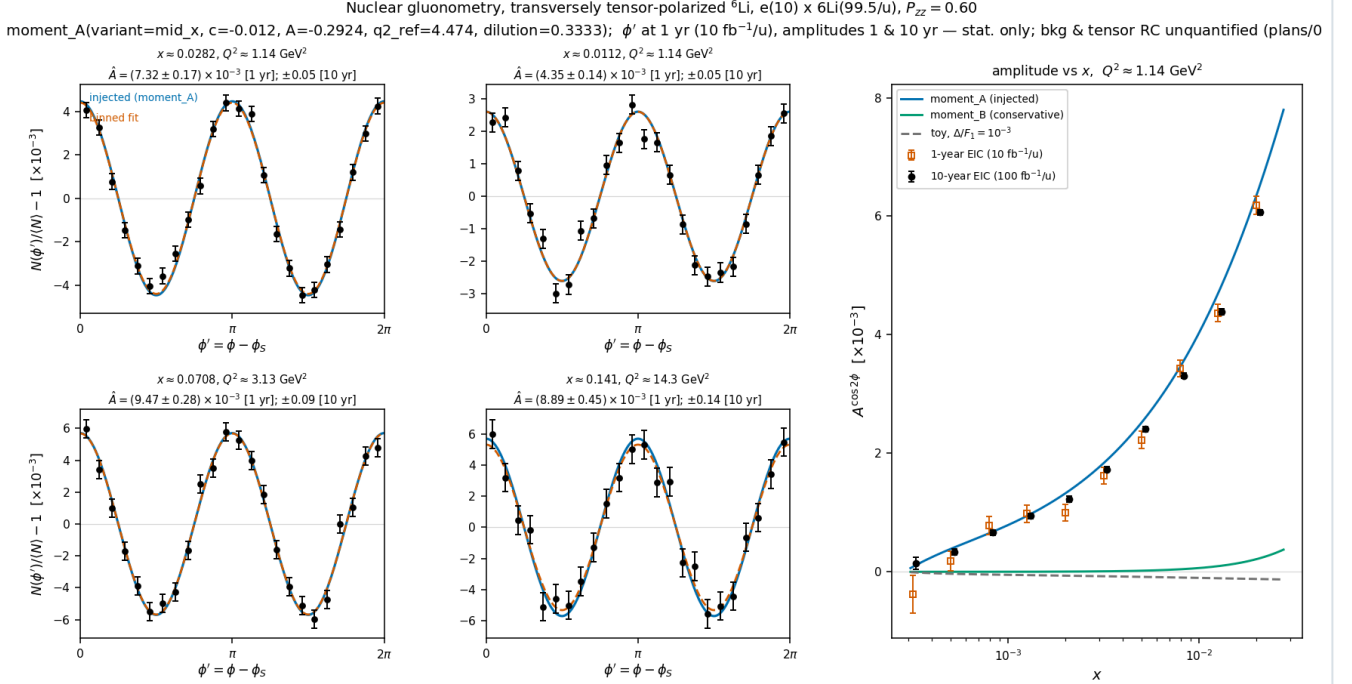
3 • The Δ model and the moment constraint

Provenance. Sather and Schmidt computed the lowest moment of Δ in the MIT bag model for the spin-3/2 Δ^{++} baryon, $\int x \Delta dx = -0.012 \alpha_s$, with its Q² evolution but no x dependence [4]; the explicit $\alpha_s(Q^2)$ is the leading-order Wilson coefficient of the photon–gluon box, so the O(α_s) suppression relative to F₁ is model-independent while −0.012 is the bag estimate. Lattice QCD adds two results, both at unphysical pion masses: the bare ϕ -meson matrix element $A_2 = 0.23(2)(5)$ at $m_\pi = 450$ MeV [25], which converts to $\int x \Delta|_\phi \approx -0.006 \alpha_s$ (our conversion, unrenormalized), and the NPLQCD deuteron δG moment of −0.010(3), bare at $m_\pi \approx 806$ MeV [27] — a transversity-to-momentum ratio roughly ten times below that of the ϕ , implying the opposite sign for the deuteron's Δ moment. Nothing exists for ⁶Li or any A > 2 nucleus beyond a $\Delta\Delta$ -isobar estimate for the deuteron [34], so the literature brackets the nuclear Δ in both directions: suppressed as in the NPLQCD deuteron, or enhanced by binding in analogy with the EMC effect [50].

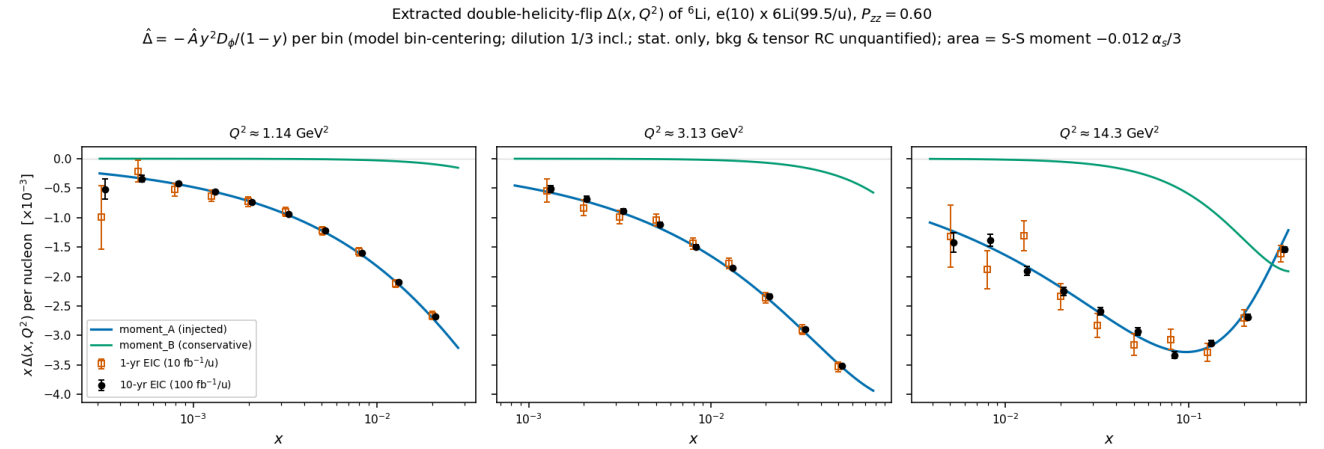
Implementation. All Δ models reside in one registry (`polli_fastsim/delta_models.py`). The production default, `moment_A`, is an ansatz normalized to the bag moment applied to the per-nucleon Δ of ⁶Li:

$$\Delta_A(x, Q^2) = \frac{1}{3} A \alpha_s(Q^2) F_1(x, Q^2) \frac{x^a(1-x)^b}{N_{\text{peak}}}, \quad \int_0^1 x \Delta(x, \langle Q^2 \rangle) dx = -0.012 \alpha_s \Rightarrow A = -0.294$$

with (a, b) = (0.7, 3); the factor 1/3 restricts the exotic glue to the polarized deuteron-like pair while the whole-nucleus P_{zz} remains a separate factor, so nothing is double counted. The constraint is imposed at the rate-weighted $\langle Q^2 \rangle = 3.9$ GeV² of the accepted phase space, and because F₁ evolves the realized moment varies between −4% and +7% across the plotted Q² slices; α_s comes from the CT18NLO table (the leading-order analytic fallback is 14% higher), and the solved $A = -0.294$ agrees with the EPPS21-based `money_delta` value of −0.31 at its $\langle Q^2 \rangle = 7.4$ GeV². `moment_B` imposes the same constraint without the F₁ factor and yields amplitudes a factor of 2–8 lower; the A-versus-B spread is the dominant ansatz systematic, and both curves appear on every plot. Porting a single-hadron bag moment to a per-nucleon nuclear structure function has no precedent in the literature, so normalization and x dependence are both scenario choices, bracketed by `moment_B` from below and possible nuclear enhancement from above.



Money plot 5 — inclusive gluonometry ($e(10) \times {}^6\text{Li}50$, $P_{zz} = 0.60$, Δ : moment_A, dilution 1/3). Left 2 $\times$ 2: ϕ' pseudo-data with 1-year statistical error bars in the four sweet-spot super-bins ($Q^2 = 1.1 \rightarrow 14 \text{ GeV}^2$, $x \approx 0.02$ -0.14); injected curve (blue), binned fit (vermillion); annotations give δA for both programs. Right: extracted amplitude vs x with 1-year (open) and 10-year (filled) error bars against the moment_A, moment_B and toy- 10^{-3} curves. Amplitudes are $(0.7$ - $1.2) \times 10^{-2}$ against $\delta A = (1.5$ - $4.5) \times 10^{-4}$ in year 1, i.e. every bin $\geq 21\sigma$, so the A-versus-B spread is resolved by the x dependence. Only the scattered electron is detected.



Money plot 7 — the extracted $\Delta(x, Q^2)$. The amplitudes of money plot 5 unfolded to the structure function (model bin-centering, $\delta \Delta = \delta A \cdot |\partial \Delta / \partial A|$), as $x \Delta$ per nucleon at the three sweet-spot Q^2 slices with independent 1-year (open) and 10-year (filled) pseudo-measurements against the moment_A and moment_B curves; the area under each curve is the ported bag moment ($\div 3$ dilution). Best bins reach $\delta \Delta \approx (0.7$ - $1.4) \times 10^{-3}$ in year 1 on $\Delta \approx -0.03 \dots -0.09$.

4 • The coherent channel

4.1 • Detecting the intact ${}^6\text{Li}$ at ePIC

In coherent diffraction, $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$, the recoil retains the beam's magnetic rigidity ($A/Z = 2$; the diffractive momentum loss $x_p < 0.01$ changes it by less than 1%) and emerges at $\theta = p_T/p \lesssim 1.7 \text{ mrad}$, inside the beam envelope. Of the far-forward system — the B0 spectrometer, the Off-Momentum Detectors (≈ 45 – 65% of beam rigidity, far-forward acceptance table of Jentsch, Tu and Weiss), the Roman Pots at $z \approx 26/28 \text{ m}$ in the Yellow Report layout (32.5/34.3 m in epic-main; $\theta < 5 \text{ mrad}$, ~ 60 – 95% rigidity for off-rigidity tracks, AC-LGAD sensors with $\sim 140 \mu\text{m}$ spatial and $\sim 30 \text{ ps}$ timing resolution) and the ZDC for neutrals [7,8,9] — only the pots can see such a recoil, and only above the 10σ envelope

cut, so at low $|t|$ it cannot be tagged at IP6 [10,51]. With an exponential coherent t -distribution the tag acceptance is analytic:

$$\varepsilon_{\text{tag}} = e^{-B p_{T,\text{cut}}^2} = 13.5\% \quad (p_{T,\text{cut}} = 0.20 \text{ GeV}, B = 50 \text{ GeV}^{-2}); \quad 4 \times 10^{-5} \quad (0.45 \text{ GeV})$$

Provenance of the cuts. The 0.20 GeV floor is the documented EIC requirement for high-acceptance (low-divergence) optics at top rigidity, a beam divergence $\leq 73 \mu\text{rad}$ for 275 GeV protons [7,48], and matches the one documented light-ion number: coherent ^3He tagging limited to $p_T > 0.20 \text{ GeV}$ at IR-6 [10]. The 0.45 GeV value is our derivation, 149 μrad high-divergence proton optics applied to the 300 GeV ^6Li momentum. No beam-optics parameters have been published for any lithium beam; with proton-class divergence at this rigidity the 10σ cut would fall at 0.5–0.7 GeV and eliminate the IP6 channel. This is the most consequential machine input to the measurement (assumption #1).

The near-beam cut is angular, and the pot aperture is measured. The 10σ envelope cuts the recoil *angle* (reconstruction-chain report §4), so the threshold scales with the beam momentum: $p_T^{\text{cut}} = 10\sigma_\theta \cdot A \cdot p_u = 0.09 / 0.22 / 0.60 \text{ GeV}$ at 40.8 / 99.5 / 137.5 GeV/u for the proton-derived $\sigma_\theta = 73 \mu\text{rad}$, i.e. tag acceptances of 67% / 9% / 1.5×10^{-8} at $B = 50 \text{ GeV}^{-2}$; the constant 0.20 GeV of the figures (13.5%) is close to the mid-energy value and far from the other two. Independently of the envelope, the pots' own aperture for an intact ^6Li has been measured by shooting one through the ePIC far-forward geometry (tools/fullsim): the boundary is $|\theta_x| \gtrsim 2.0 / 1.35 / 1.03 \text{ mrad}$ in the $5 \times 41 / 10 \times 100 / 18 \times 275$ optics against $|\theta_y| \gtrsim 1.8\text{--}3 \text{ mrad}$, so the acceptance opens **horizontally**, where the chain's assumed cutout — wide in x , tight in y — opened it vertically; the cause is the transport, which images an IP angle onto the pot plane with a horizontal lever $R_{12} \approx 30.6 \text{ m}$ against a vertical few metres. A recoil must clear both constraints and the aperture is the larger everywhere. Carried through the chain at the low configuration: acceptance $37.7\% \rightarrow 1.42\%$, $N_{\text{tag}} 8.3 \times 10^6 \rightarrow 3.1 \times 10^5$, acceptance-induced $\langle \cos 2\beta \rangle +0.426 \rightarrow -0.772$, a change of sign; two of the four $|t|$ bins survive ($|t| = 0.061\text{--}0.273 \text{ GeV}^2$) with δa_t worse by $6\text{--}34\times$ ($0.0014 \rightarrow 0.0482$, $0.0026 \rightarrow 0.0150$), and a_e is still recovered at 0.0073 ± 0.0045 and 0.0091 ± 0.0045 against an injected 0.0100. At mid and top the aperture leaves no accepted recoil in the binned window: the coherent program is a low-energy program. The scan is one event per point in 30° azimuthal steps, without a beam envelope, on a September-2024 epic-main geometry (plans/04 #20). The coherent figures below predate the measurement and keep the constant 0.20 GeV cut; money plot 6R is on the angular envelope and that assumed cutout ($|p_x| < 0.55$, $|p_y| < 0.22 \text{ GeV}$, with divergence smearing): 2.7×10^6 tagged events at mid energy in year 1, a fake $\langle \cos 2\beta \rangle = +0.77$ cancelled by the ratio, $a_t(t_{\text{ref}}) = 0.117 \pm 0.007$ for 0.119 injected and $a_e = 0.0101 \pm 0.0013$ for 0.010, for a background-free, Z-identified sample by assumption (the breakup budget is plans/06).

Coherence selection at IP6 and at IR-8. Standard EIC practice at IP6 establishes coherence by vetoing breakup rather than by tagging the recoil (§5). Direct low- $|t|$ tagging is a central argument for the second interaction region, whose secondary beam focus $\sim 45 \text{ m}$ downstream ($\sigma \approx 0.15 \times 0.1 \text{ mm}$) lets Roman Pots at $z = 44.0/45.5 \text{ m}$ reach $p_T \approx 0$, with published intact-recoil efficiencies of 47% (d), 32% (^3He), 29% (^4He) and 17.8% (^7Li) at top energy, rising steeply at lower energies (^3He reaches 99.8% at 5×41) [10]; no ^6Li case is published and our interpolation is $\approx 20\%$. One degeneracy remains even there, since a Roman-Pot hit measures only A/Z and intact ^6Li , α and d at the same velocity share trajectory and timing: the second-detector design answers it with a proposed Z^2 -scaling Cherenkov counter behind the pots, at the R&D stage [8], while for the IP6 AC-LGADs no charge-discrimination study exists (amplitude is used only for position interpolation) and the minimum-angle ion acceptance is not fully established [49]. That question, open item #19, decides whether IP6 coherent-tag purity is established event by event or statistically (§5).

4.2 • Rate model

$$\sigma_{\text{coh}} = f_{\text{coh}}(x) \sigma_{\text{DIS}}, \quad f_{\text{coh}}(x) = \frac{f_0}{1 + (x/x_{\text{coh}})^2}, \quad f_0 = 0.04 \quad [0.02\text{--}0.08]$$

No diffractive prediction exists for a light nucleus (the lightest published case is Ca), so the f_0 band brackets HERA ep diffraction (10–15% of DIS [44]) and heavy-nucleus saturation estimates (20–25% [45]); parameterizing f_{coh} in Bjorken x rather than x_p is optimistic for high-mass diffraction, which that band is meant to cover. The scale $x_{\text{coh}} = 0.01$ marks where the coherence length $l_c = 0.105 \text{ fm}/x_p$ falls below the ^6Li size, and the slope $B = 50 \text{ GeV}^{-2}$ (band 40–60) follows from $B = \langle r^2 \rangle / 3$ with the matter radius (the charge radius gives ≈ 57 [47]; gluonic distributions may be wider [42]), valid

below the form-factor minimum at $|t| \approx 0.31 \text{ GeV}^2$ [11,12]. The yields are $1.1 \times 10^7 / 1.1 \times 10^8$ tagged events in the 1- and 10-year programs, best-super-bin $\delta A = 1.9 \times 10^{-3} / 6 \times 10^{-4}$.

4.3 • The anchored amplitude

The $\cos 2\phi'$ modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron CGC calculation of Mäntysaari, Salazar, Schenke, Shen and Zhao [13], the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's subsequent work is unpolarized [14,15]). Their coefficients, digitized in

`coherent.MANTYSAARI_A2_DEUTERON`, show a_2 growing linearly in $|t|$ with $a_2(m=0) = -2a_2(m=\pm 1)$, from the 6% D-wave modulating the diffractive slope. For a fill with populations p_m the ensemble coefficient is proportional to P_{zz} :

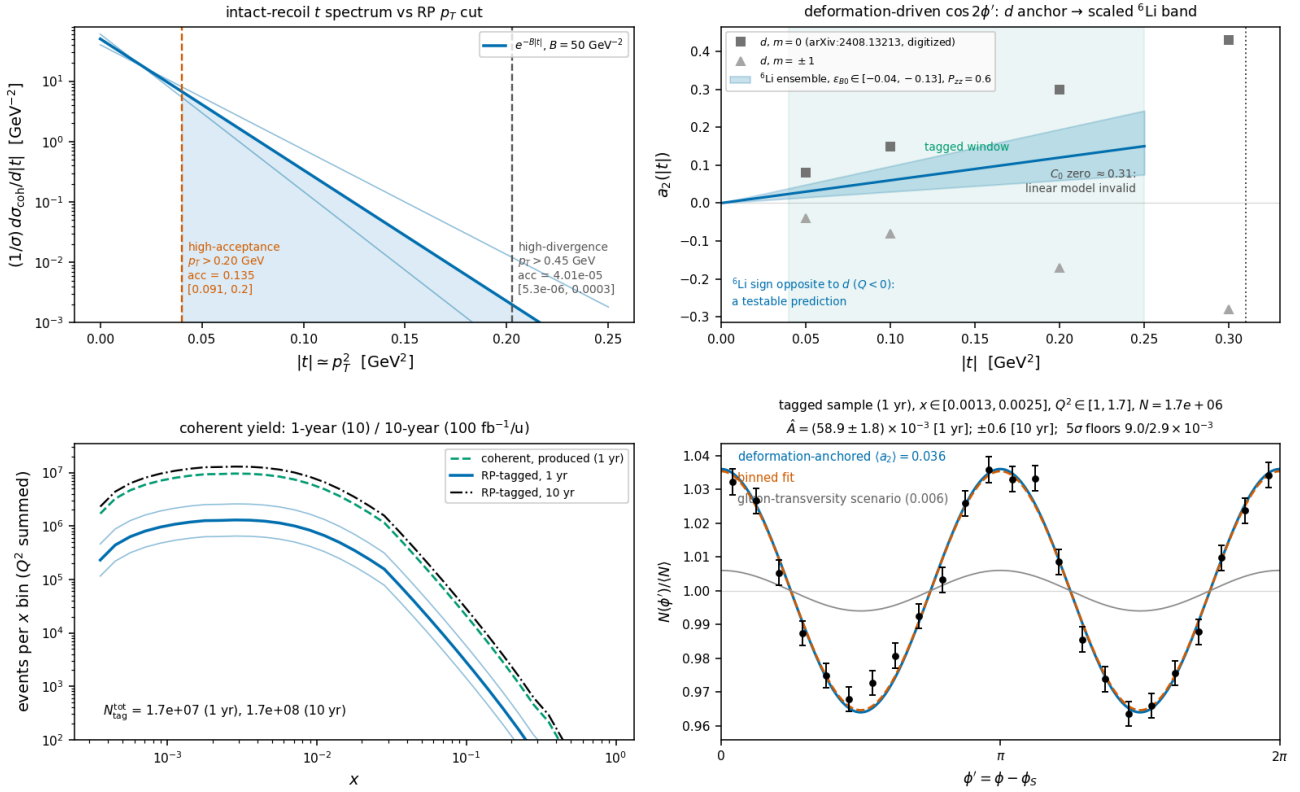
$$a_2(t; P_{zz}) = -\frac{P_{zz}}{4} \varepsilon_{B0} B |t|, \quad \langle a_2 \rangle_{\text{tag}} = a_2(p_{T,\text{cut}}^2 + 1/B) = 0.036 \quad [0.018-0.059]$$

Convention of the anchored coefficient. The anchor's Eq. (9) expands $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ with Φ the azimuth of the produced vector meson — the momentum transfer, i.e. the recoil — relative to the polarization axis in the γ^*d frame, so the $\cos 2\Phi$ coefficient is $2a_2$: the $\langle a_2 \rangle$ quoted here and injected in money plot 6(d) understates the deformation modulation by a factor two (0.072 rather than 0.036 at $P_{zz} = 0.6$), and this report's coherent numbers are conservative by $\times 2$ and kept in the a_2 convention. The modulation is one of the *recoil* azimuth $\phi_t - \phi_S$, not of the electron azimuth used as the axis of panel (d); plot 6R injects $2a_2$ in the recoil azimuth and fits both azimuths (reconstruction-chain report §7).

$\varepsilon_{B0} \equiv \Delta B_0/B$ is scaled from the deuteron value of 0.21 by two indicators that disagree: the relative quadrupole deformation $Q/\langle r^2 \rangle$ of ${}^6\text{Li}$ is about one fifth of the deuteron's and opposite in sign, while the α -d asymptotic D/S ratio is disputed — a transfer measurement gives $\eta = +0.0003(9)$, consistent with zero [16], against older sub-Coulomb analyses favoring $\eta \approx -0.01$. We adopt $\varepsilon_{B0} \in -(0.04-0.13)$ with default -0.08 , leaving near-zero deformation open; the 5σ floors cover its detection or exclusion. The band's width is genuine theoretical uncertainty: even exact few-body calculations miss the ${}^6\text{Li}$ quadrupole cancellation by a factor of ≈ 3 [17], and the gluonic deformation need not share the cancellation seen in the charge sector. The sign flip relative to the deuteron is a testable prediction ($Q({}^6\text{Li}) < 0$), and tensor modulations of comparable size are established at these $|t|$: e-d $T_{20} = -(0.1-0.3)$ [19] and HERMES $A_{zz}(d) \approx -1\%$ at $x \approx 0.01$ [20], the latter attributed to coherent double scattering [21–23]. The flat gluon-transversity term is separately bounded at $3 \times 10^{-3} - 10^{-2}$, from the bare lattice ϕ transversity [25] divided by the $\sim 10\times$ NPLQCD deuteron suppression [27], with the maximal Drell–Yan scenario [5] as the ceiling. The equal-rate population average behind $\langle a_2 \rangle$ carries a one-sided $\times 0.73$ rate-weighting systematic, inside the ε_{B0} band.

Separation of the two mechanisms. The two differ in shape: deformation is linear in t , sign-flipped relative to the deuteron, and expected to weaken toward small x_P (JIMWLK evolution washes it out [15]), while exotic glue is flat in t and persists at small x , where it connects to the elliptic-gluon program [6]. In a two-component fit $a_2(t) = c_{\text{def}} \cdot |t| + a_g$ per x_P bin, both components exceed the statistical floor by $\geq 5\sigma$ at $100 \text{ fb}^{-1}/u$. Incoherent breakup is m-state-blind in this mechanism [13]: it dilutes the modulation but cannot mimic it.

Coherent $e^6\text{Li} \rightarrow e'X^6\text{Li}(\text{g.s.})$, transversely tensor-polarized, $e(10) \times 6\text{Li}(99.5/u)$, $P_{zz} = 0.60$
 nario rates; modulation anchored on the polarized- d CGC calculation, PLB 858:139053 -- plans/06 SS6.4b; statistical only, backgrounds/purity and tensor RC unquanti



Money plot 6 — coherent channel, $e^6\text{Li} \rightarrow e'X^6\text{Li}(\text{g.s.})$ (same beam, $P_{zz} = 0.60$). Top left: recoil t -spectrum vs both optics' p_T cuts with the slope band. Top right: digitized deuteron a_2 per m -state [13] vs the scaled ^6Li ensemble (sign flip predicted), with the tagged window and the form-factor-zero exclusion marked. Bottom left: tagged yield vs x for both programs with the f_0 band ($N_{\text{tag}} \approx 1.1 \times 10^7 / 1.1 \times 10^8$). Bottom right: ϕ' pseudo-data (1-year statistics) at the anchored (a_2) = 0.036 with the flat gluon-transversity scenario (gray); extraction $\hat{A} \pm 1.9 \times 10^{-3}$ (1 yr), $\pm 0.6 \times 10^{-3}$ (10 yr).

5 • Backgrounds for the coherent tag (full budget: plans/06)

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
$\alpha + d$ breakup (threshold 1.474 MeV; the 2.186 MeV 3^+ state, $\Gamma = 24 \text{ keV}$, decays $\sim 100\%$ $\alpha + d$ [39])	Both fragments at $R = 1.000$, on the trajectory and velocity of an intact ^6Li : tracking and ToF blind; only dE/dx ($\propto Z^2$: 9/4/1) separates	RP charge discrimination — open question #19 : no study exists for the IP6 AC-LGADs (the documented Z-ID is the proposed IR-8 Cherenkov [8]). Fallback: two-component $ t $ fit (coherent $e^{-50 t }$ vs $\sim 10\times$ flatter incoherent; crossover at 0.05–0.2 GeV 2 inside the window [41,42]); e+Pb benchmark: 80–99% incoherent rejection [43]
Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), $^3\text{He} + t$ (15.79 MeV) [39]	A fragment could pass unnoticed	$p \rightarrow \text{OMD}$ ($R = 0.50$), $n \rightarrow \text{ZDC}$, $^3\text{He} \rightarrow \text{RP window}$ ($R = 0.75$) [51]: vetoable; inefficiency is second-order
$^6\text{Li}^*$ 3.5629 MeV (0^+ , $T = 1$, $\Gamma = 8.2 \text{ eV}$) [39]	Particle-stable ($\alpha + d$ parity-forbidden, $\Gamma_{\alpha} \leq 6 \times 10^{-7} \text{ eV}$); γ -decays to a real $A = 6$, $Z = 3$ track	Isoscalar exchange cannot drive $T = 0 \rightarrow 1$; the boosted 0.1–0.4 GeV γ lands mostly in the B0 calorimeter (photons $\geq 50 \text{ MeV}$, 5.5–20 mrad [9]), tail in the ZDC, and the 5.366 MeV $2^+ T = 1$ state always emits a nucleon. Every $T = 1$ state is γ - or nucleon-vetoable; only $T = 0$ states feed the blind $\alpha + d$ channel
Bethe–Heitler / QED Compton	Intact nucleus by construction	Calculable (t -dependence = charge FF^2 [11,12]); subtract and reuse as an RP-acceptance candle; the charge-quadrupole $\cos 2\phi$ is $< 10^{-3}$ for $ t < 0.15 \text{ GeV}^2$ [12,17]
Beam-gas / halo; e' -side fakes	Accidental $R \approx 1$ tracks; $\pi /$ pair-symmetric e' at high y	Non-colliding bunches, $\sim 30 \text{ ps}$ timing, vertex association; standard DIS e' ID plus (e' , recoil) kinematic consistency

Two extraction systematics survive the spin-state ratio of §2: the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$, common with A_{zz} , and radiative corrections for tensor observables, which have not been calculated and are noted on every plot. In interpretation, a $\Delta\Delta$ -isobar admixture can mimic Δ at leading twist [34] and conventional double scattering generates the low- x b_1 [21–23]; the Q^2 lever arm and the ${}^6\text{Li}/d$ comparison discriminate against both.

The ${}^6\text{Li}$ null test. Deformation-driven coherent modulations, the polarized-deuteron imaging signal [13], are bounded on ${}^6\text{Li}$ by its anomalously small quadrupole moment, $Q = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$ [35]: a sizable flat-in- $t \cos 2\phi$ on ${}^6\text{Li}$ cannot be a shape effect, and the mechanisms entangled for the deuteron separate in the ${}^6\text{Li}/d$ comparison.

6 • Critical assumptions

#	ASSUMPTION	IMPACT IF WRONG
1	The Roman-Pot p_T cut and the measured pot aperture (§4.1)	Tag acceptance from 13.5% to negligible at IP6; the case moves to the low-energy configuration and to IR-8, whose secondary focus removes the cut entirely [10]
2	Coherent fraction $f_0 = 0.04$ [$\times 2/\div 2$], in Bjorken x (§4.2); no diffractive model for a light nucleus exists (#18)	Yields scale linearly; δA as $1/f_0$
3	Bag moment ported to per-nucleon ${}^6\text{Li}$ with 1/3 dilution; x -shape ours (§3)	Overall Δ scale; the literature brackets both directions [27,50]
4	ε_{B0} scaled from the deuteron by two dials disagreeing by $\times 3$ (§4.3)	Deformation amplitude anywhere in the band, which every plot carries; the sign flip is robust
5	Event-by-event Z-ID in the pots is NOT assumed; purity from the $ t $ -shape fit (§5, #19)	Z-ID would improve purity; an incoherent slope model that is off grows the fit systematic
6	Linear $a_2(t)$ quoted only for $ t < 0.2 \text{ GeV}^2$; the form-factor zero at 0.31 GeV^2 is excluded, not modeled	The upper tagged window carries unmodeled structure
7	Toy structure functions with grid backends, $R = \sigma_L/\sigma_T$ included; no target-mass terms; $g_2 = g_2^{WW}$ (§1)	Rates to $\times 1.5$; shapes are scenarios; the published R moves Δ/F_1 by -4 to $+18\%$
8	All luminosity in transverse-tensor fills at the in-ring placeholder $P_{zz} = 0.6$ (sources reach $\sim 95\%$ of maximal P_{zz} [37]); the $+0.6 / -1.2$ pattern shares it and is $1.5\times$ more precise	N scales with the run-plan share; the amplitude with P_{zz}
9	Impulse approximation; no FSI; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger $ t $; RC unquantified, though the QED tail is calculable and doubles as calibration
10	Equal-rate population average for $\langle a_2 \rangle$; t_{min} neglected in the tag acceptance	One-sided -27% (rate weighting) and $\approx -10\%$ (t_{min}) — inside the ε_{B0} / f_0 bands

7 • Open inputs

The simulation chain's remaining gaps are plans/08 and the external dependencies plans/04. The highest-leverage single input is a rerun of the IP-Glasma polarized-deuteron setup [13] with an α - d cluster density, which would replace both ε_{B0} indicators and the f_0 band at once; a lattice Δ moment for a light nucleus, or a measured tensor form factor read through the sum rule tying it to transversity-GPD moments [30], would pin the normalization of §3.

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Appendix A · Modification history

DATE	REVISION
2026-08-10	First version (development run 5): money plots 5 and 6 with the moment-constrained $\Delta(x,Q^2)$ and separate 1-year / 10-year projections, and the physics note with a verified bibliography (docs/note_cos2phi_coherent_6Li.md); first audit round the same day.
2026-08-11	Second audit round (formulas against the literature, implementation, references, extraction logic); money plot 7, the extracted Δ as structure-function data points, added.
2026-08-17	Dedicated fact checks of the Sather-Schmidt moment usage (a Δ^{++} bag calculation without x-shape) and of the far-forward detection chain; phase-space figure with the analysis binning; display mathematics typeset at build time.
2026-08-24	The anchored coefficient's convention (§4.3): the anchor's Eq. (9) expands $1 + 2\sum a_n e^{in\Phi}$, so the $\cos 2\Phi$ coefficient is $2a_2$ and this report's coherent numbers are conservative by $\times 2$; the reconstruction-chain report created as the reconstructed-level companion.
2026-08-25	The spin-state-sorted ratio replaces the single-state fit as the estimator of the measurement, with the 5R/7R results quoted; the near-beam cut stated as angular, with the slot-like cutout and the 6R numbers; the b_1 , b_2 sign convention; the inversion corrected to $\Delta = -\hat{A} y^2 D_\phi / (1-y)$; the sweet-spot y^2 statement corrected, with the $O(y^2)$ b_1

leakage; far-forward parameters (OMD window, pot z positions, high-divergence figure) updated; reference [51] flagged incomplete; test count 94 + 24. Figures and projections unchanged. Modification history moved to this appendix.

2026-08-26	Development run 8: the hadronic final state is a PYTHIA 8 sample rather than a Gaussian stand-in (§2); the Roman-Pot aperture for an intact ⁶ Li is measured in the ePIC geometry and the coherent chain re-run on it (§4.1); the published R1998 fit is transcribed and threaded behind an r_func hook, with its effect on Δ/F ₁ measured (§1); bin-centering has a forward-folded alternative (§2). Work planning moved to plans/08 and plans/04; figures unchanged.
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Generated from the PolarizedLithiumSim repository: `evgen/scripts/money_cos2phi.py` , `evgen/scripts/money_cos2phi_coherent.py` , `evgen/scripts/money_delta_extraction.py` , `evgen/scripts/phase_space_bins.py` ; models in `polli_fastsim/delta_models.py` and `evgen/polligen/coherent.py` ; assembled by `reports/build_report.py` (display math typeset at build time, matplotlib mathtext/STIX). 183 evgen + 48 fastsim tests pass; every command is in docs/reproduction_manual.md. All rates use TOY/scenario inputs — bands, not predictions. External numbers are verified against primary sources (nuclear data: TUNL; detector: EIC YR, ATHENA, ePIC and second-IR documents; theory: original papers). Target venue: INT program on polarized ion beams at the EIC, March 22 – April 2, 2027.